BRANDON REISS

Summit, NJ (NYC metro) | brandonreiss.com | LinkedIn

SUMMARY

Applied AI systems architect and former founder specializing in production ML platforms that handle real-world scale, latency, and cost constraints. Proven track record of operating autonomously to translate ambiguous R&D bets into durable production systems, architecting high-throughput unstructured data intelligence layers, and advising executive leadership on company-wide AI strategy.

EXPERIENCE

HubSpot | Remote
Jan 2025 – Present

Staff Software Engineer

- Architect and top contributor for HubSpot's unstructured data intelligence layer processing ~200B tokens per day and ~30M globally distributed record update triggers (>600 RPS peak) across 60M orchestrated extraction tasks, while maintaining p99 end-to-end latency under 4 minutes and unit costs below $0.001 per record.
- Integrated unstructured context into flagship AI assistant, setting evaluation standards from experimentation to release; achieved >30% daily traffic penetration with measurable lifts in CSAT and model quality benchmarks.
- Designed company-wide unstructured content normalization and noise-filtering standards, significantly cutting inference token waste, reducing hallucinations, and establishing sustainable unit economics for LLM features.
- Advise cross-organizational teams on AI platform standards, focusing on model evaluation methodology, inference cost optimization, and low-latency architectural reliability.

Frame AI | New York, NY
Feb 2016 – Jan 2025 *(Acquired by HubSpot)*

Co-Founder / Principal Engineer / Director

- Co-founded and served as principal architect from inception to acquisition by HubSpot, scaling an applied AI platform over 4 orders of magnitude to handle millions of monthly conversations powering dozens of process automations at a dozen mid-market and enterprise clients.
- Provided technical leadership to lean engineering organization of 6 across full-stack, data engineering, R&D, and forward-deployed functions, delivering major yearly product releases, weekly customizations, and a major white-label product integration with Twilio Flex.
- Architected multi-stage ingestion, extraction, and synthesis pipelines processing continuous streams of customer interactions under demanding end-to-end SLAs.
- Implemented task prioritization and deduplication strategies across heterogeneous computational workloads and multi-tenant fault isolation across service fleet.
- Synthesized classical NLP and modern ML into configurable, cost-efficient production pipelines—unifying rules-based filtering, dependency parsing, sentence encoders, and LLMs into cohesive intelligence systems.
- Engineered storage backends combining relational databases (PostgreSQL) for transactional state with columnar and distributed object/NoSQL storage (ClickHouse) for high-volume analytical workloads.
- As Head of Security, successfully unlocked Fortune 100 go-to-market strategy by establishing comprehensive security, compliance, and risk management programs; achieved and maintained SOC 2 Type II certification for 5 consecutive years with zero notable exceptions, delivered HIPAA compliance, and cleared security audits.

Knewton | New York, NY
Sep 2012 – Dec 2015

Software Engineering Manager / Senior Engineer

- Selected by senior leadership to lead a high-performing team of 8 engineers within the Adaptive Learning division, bridging core R&D with production engineering and setting technical execution standards.
- Built low-latency (p99 < 2 minutes) statistical inference engines and recommendation systems serving millions of students globally.

Organic Motion | New York, NY — Jan 2007 – Sep 2012
Software Engineer

- Developed low-latency computer vision algorithms and local network distributed systems for markerless motion capture and clinical gait analysis.
- Self-taught hardware and firmware development to produce custom PCBs mounted with rings of near-IR LEDs and programmable microcontroller.

TECHNICAL EXPERTISE & CORE COMPETENCIES

Agentic AI & Compound Systems → Token Budgeting & Context Window Management, Tool & Skill Architectures, Agent Task Performance Evaluation & Benchmarking, Task Aligned Data Retrieval, RAG.

Applied AI / ML & Statistical Inference → Recommender & Ranking Systems, Multi-Class Text Classification, Generalized Linear Models (GLMs), Signal Extraction & Feature Engineering, Sentence Encoders, Clustering, Syntactic/Dependency Parsing, Classical NLP & Rule-Based Heuristics.

Distributed Systems & Concurrency → Low-Latency Event-Driven Architectures, Multi-Tenant Resiliency (Blast-Radius Isolation, Anti-Correlated Failure Design), Priority-Tiered Processing, Fault-Tolerant Distributed Queues & Worker Architectures, Asynchronous / Parallel / Concurrent Execution Patterns, CPU & Memory Profiling.

Data Infrastructure & Storage → Relational Databases (PostgreSQL & InnoDB Internals, Concurrency, Lock Mechanics, Indexing), Columnar OLAP Datastores, Distributed Storage (HBase, S3).

Security, Compliance & Governance → SOC 2 Type II Program Architecture & Audit Defense, HIPAA Compliance, Enterprise Security Reviews, Risk Management & Data Governance, Application Security Testing.

Languages & Technical Tooling → Python, C++, Java, SQL, Bash / Linux CLI Tools; broad polyglot proficiency across modern software stacks; agentic coding harnesses & AI-assisted development workflows.

EARLY CAREER & APPLIED RESEARCH

Undergraduate Research Assistant — Rensselaer Polytechnic Institute (RPI)
Co-authored peer-reviewed research in Applied and Environmental Microbiology (April 2006, 72: 2460–2470) on isolating PCB-dechlorinating bacterial cultures using PCR, DNA sequence analysis, and gas chromatography.

Engineering Intern — BAE Systems
First high-school intern admitted in company history; designed a $300K RF-absorbent antenna test facility and conducted experimental classification of RF signal polarization using neural networks.

EDUCATION

M.S. in Computer Science — New York University, Courant Institute of Mathematical Sciences
Recipient, Courant Hedwig Kurzbart Master's Fellowship (awarded annually for academic excellence)

B.A. in Computer Science and Mathematics — New York University, College of Arts and Science
Minor in Metropolitan Studies